
RUJUL SHARMA

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SUMMARY

Design Verification Engineer with hands-on experience in **UVM-based verification, SystemVerilog testbench architecture**, functional coverage, and assertions. Background includes verifying communication protocols (SPI, APB), automating flows, and debugging complex SoC designs. Passionate about building robust and scalable verification environments.

EDUCATION

Arizona State University, Tempe

MSE Electrical Engineering

Aug 2021 – May 2023

GPA-3.07/4.00

Graduate Coursework: Digital Systems and Circuits, Python for Rapid Engineering, Analog Integrated Circuits, VLSI Design, Computer Architecture, Constructionist Approach to Microprocessor Design.

Manipal University, Jaipur

B.TECH. in Electronics & Communication

July 2017 – May 2021

GPA- 8.00/10.00

TECHNICAL SKILLS

Programming & Scripting Languages: SystemVerilog, UVM (Universal Verification Methodology), Functional Coverage, System Verilog Assertions (SVA), Object-Oriented Programming (OOP), C, Python, TCL, Perl.

Tools and Technologies: Cadence Xcelium, Hspice, Quartus Prime, ModelSim, APR, Cadence Virtuoso, Arduino, MATLAB, CST, Linux.

PROJECTS

Verification of APB_RAM

Jan 2024 – Feb 2024

- Orchestrated the creation of a modular UVM-based verification environment for the APB RAM enabling structured and scalable verification with seamless component integration and achieved 100% functional coverage in alignment with verification goals.

Verification of Serial Peripheral Interface (SPI)

Sep 2023 – Nov 2023

- Constructed hierarchical and reusable UVM-based SystemVerilog testbenches for the SPI module, enhancing efficiency and maintainability.
- Designed UVM components like sequence items, sequencer, driver, monitor, scoreboard to rigorously validate design correctness and performance.
- Implemented both directed and random constraint test cases, systematically achieved 100% functional coverage, and incorporated assertions to fulfill verification objectives outlined in the test plan.

Bit2Netlist

Jul 2022 – Dec 2022

- Conducted extensive full chip regression tests in Xcelium, resolving errors and ensuring robust functionality of the generated core sector.

RTL to GDSII Design of a 2-bit adder (ASU/ARM's 7nm FinFET PDK)

Spring 2022

- Created a 2-bit adder module using SystemVerilog and confirmed the functionality using ModelSim.
- Synthesized the design using Design Compiler and carried out DRC and LVS checks for the design.

ASIC Acceleration for Graph Convolution Neural Network (7 nm Tech)

Spring 2022

- Constructed a GCN module in SystemVerilog, leveraging Feature Aggregation and Transformation techniques.
- Validated the design using a predefined testbench in ModelSim, ensuring functionality, and performed synthesis in DC compiler to address timing issues.

1X3 Router Design

Mar 2021 – Apr 2021

- Formulated a 1x3 Router using Verilog HDL, modeling multiple modules (FSM, Register, Synchronizer, FIFO) and integrated the TOP module.
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WORK EXPERIENCE

Syndicate-Services LLC

Apr 2024 – Present

QA Automation Engineer

- Devised Python-based test plans and automated testing scripts to identify bugs and enhance performance in software application, ensuring comprehensive test coverage and improved efficiency.

Epics Program, Arizona State University, Tempe

Aug 2023 – Apr 2024

Volunteer

- Worked alongside faculty on projects involving combinational logic design, communication interfaces (SPI, I2C, UART) and Bus Protocols (APB, AXI) and developed UVM-based testbenches to verify functionality.

Lattice Semiconductor Corporation, San Jose

Jul 2022 – Dec 2022

Design Verification Intern

- Automated Bit2netlist flow using Makefile scripting, reducing manual intervention and improving integration efficiency and reduced simulation time by 90% and verified 20+ testcases.
- Played an integral role in the full-chip verification process by debugging and actively participated in the development of an Automation Script tailored specifically for a Bit2netlist flow.
- Collaborated closely with cross-functional design and software teams to debug and resolve issues/errors while enhancing and rectifying pre-existing test cases.

Maven Silicon, Bangalore

Jan 2021 – Jun 2021

Design Verification Intern

- Designed and implemented digital systems using Verilog/SystemVerilog, executing projects such as FSMs, FIFOs, and a vending machine while applying verification methodologies to ensure robust and efficient functionality.
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ASSESSMENTS AND CERTIFICATIONS

- Udemy- Verification Series: UVM Essentials.
 - Udemy- Verification Series UVM Projects.
 - Udemy- SystemVerilog for Verification.
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